

Caravan Policy Ownership in the COIL 2000 Insurance Dataset

A Reproducible Data Workflow

Ronald B.

Udacity Institute of AI and Technology

AI Programming Foundations Project

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Overview

In this project I built a reproducible data workflow in a Jupyter notebook. For the data I have used the COIL 2000 dataset from UCI Machine Learning Repository <https://archive.ics.uci.edu/dataset/125/insurance+company+benchmark+coil+2000>. The dataset contains real customer data from a Dutch insurance company (van der Putten, 2000). The notebook loads and cleans the data, has a reusable EDA function to explore the data, includes three visualizations and has a summary of the findings at the end. The main question was which customer characteristics are linked to having a caravan insurance policy.

Dataset Description

The data represents real customers of a Dutch insurance company, collected for the CoIL Challenge 2000. It contains 5,822 customers and 86 columns with no missing values. There are two kinds of columns: columns 1 to 43 describe the postal code area that the customer lives in and columns 44 to 86 describe the customer's own insurance products. Van der Putten and van Someren (2004) explain that the area data is linked to the postal code and not to the individual customer. The target column is CARAVAN, this information tells if the customer has a caravan policy or not and only 6% of the customers have one. I focused on a few key variables like the number of car policies, purchasing power class, customer main type and average age band. The UCI download also includes a separate file with 4,000 extra customers without the caravan column, which I did not use in this project.

Workflow Description

For ingestion I loaded the tab-separated file, which has no header row, took the 86 column names from the dictionary.txt file and ran first quality checks for missing values, duplicates and summary statistics. To clean the dataset I made two functions, one to rename the columns to readable English and one to turn the value codes into labels, both returning a new DataFrame so the original data remains the same. For the exploratory analysis I made a `caravan_rate_by_group` function that I reused on four characteristics. I added three bar charts for the visualizations with a 6% average as a reference line. To conclude the notebook, I added a summary with my findings, limitations and open questions. I added a short Markdown note after each step, because keeping code and explanation together in one notebook supports a transparent and reproducible workflow (Danchev, 2022).

Key Decisions and Assumptions

One of the cleaning choices I made was renaming and decoding, because a code like `MOSHOOFD = 7` is not informative to a reader. Another cleaning choice was keeping the 602 duplicates, because identical rows can be real customers from the same postal code. My focus for EDA was caravan ownership. I chose this focus because it was the original question of the CoIL Challenge (van der Putten & van Someren, 2004). In the EDA I compared each group against the overall 6% average. Because the groups with a small number of customers can skew the results, I left them out of Figure 1, such as the 19 customers with 3 or more car policies. I made three visualizations. Figure 1 is about the customer's own products, showing the strongest link. Figure 2 is about the type of area. Figure 3 keeps the classes in order so it clearly shows the rise for each higher class and the drop at class 8. The first assumption I made was that the order of classes starts at the lowest 1 and ends at the highest class 8. To support this assumption, I checked it

against income and home ownership. Furthermore, I assumed that the area data stands in for the individual, because there was no individual information in the dataset to use. Lastly, I assumed that having a caravan policy means the customer also owns a caravan.

Results and Interpretation

Figure 1 shows 2.5% with no car policy, 8.7% with one car policy and 15.4% with two car policies, the last one far above the average of 6%.

Figure 1

Caravan Rate by Number of Car Policies

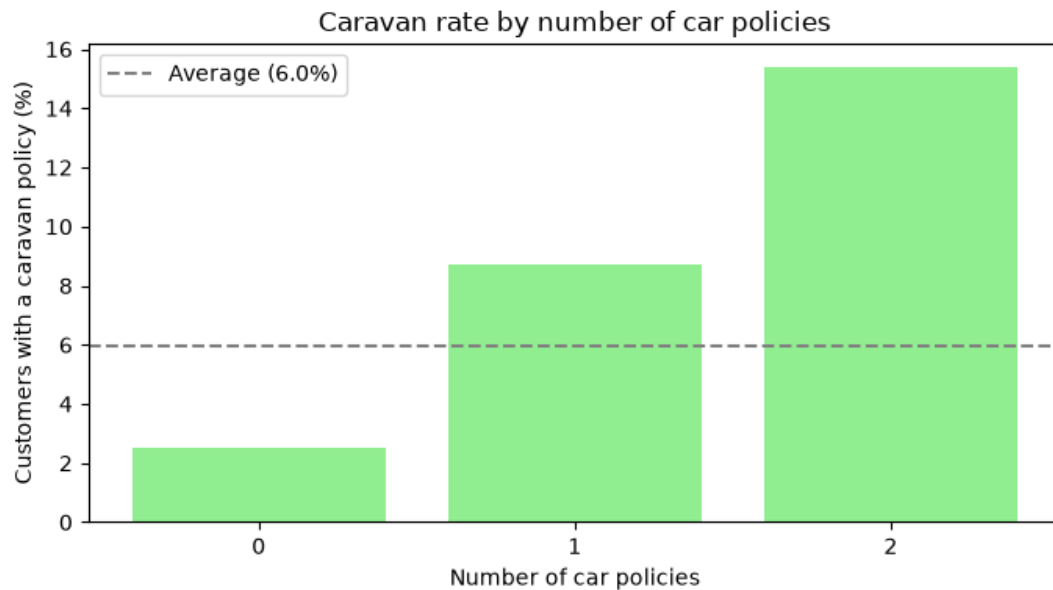


Figure 2 is about the customer main type of the area. Driven growers are the highest with 13.1% and Successful hedonists at 8.7%. The lowest are Farmers with 1.8%, Cruising seniors 2.0% and Career loners at 0%, but there are only 52 customers in that group.

Figure 2

Caravan Rate by Customer Main Type of the Area

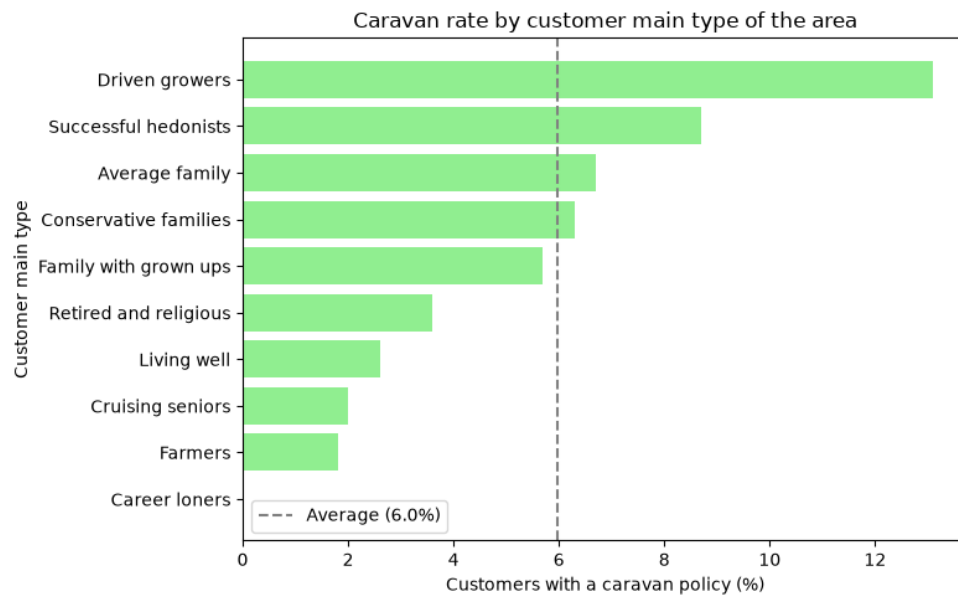
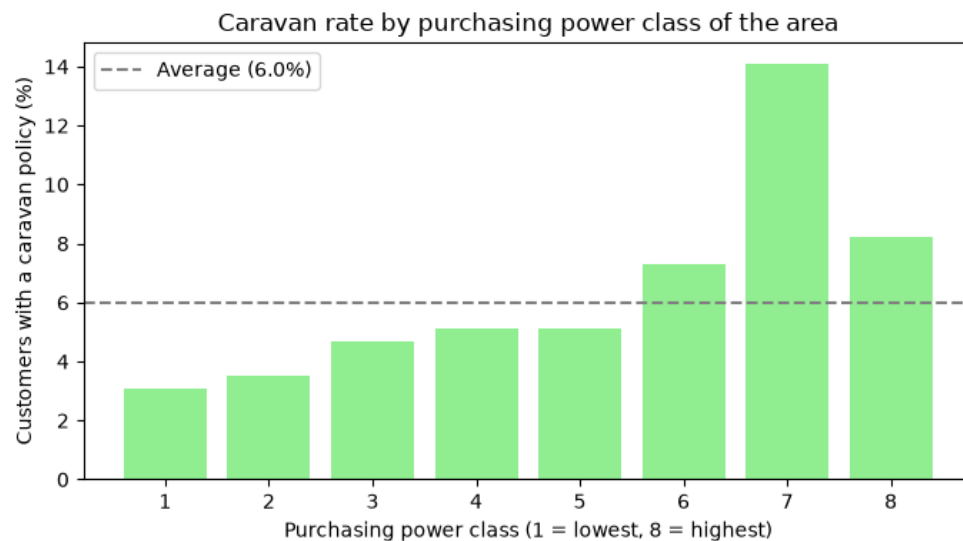


Figure 3 shows the purchasing power classes. It rises from 3.1% in class 1 to 14.1% in class 7, then drops to 8.2% in class 8.

Figure 3

Caravan Rate by Purchasing Power Class of the Area



I also checked the average age band of the area, but the rates barely differ from the average of 6%, so age does not seem to be linked to having a caravan policy. A likely explanation for Figures 1 and 3 is that owning a caravan requires a car to tow it and money to buy one. If you look at it from a business angle, existing car customers are the most promising group to offer a caravan policy. Another interpretation is that the higher the purchasing power class, the more likely it is that people have a caravan policy. This relationship holds up until class 7, but not for the highest class 8. These results match van der Putten and van Someren (2004), who found that the number of car policies and the purchasing power class were among the most predictive features in this dataset and that their relation with caravan ownership mostly went up step by step, but not always.

Responsible Practice (Bias and Data Quality)

Dropping duplicate rows would not remove rows at random. Identical rows mostly come from customers in the same area with the same products, so dropping them would reduce some customer types more than others. That is why I kept them in the dataset. Small groups give unreliable percentages, so I reported the group sizes and left the smallest group out of Figure 1. Because most columns describe the postal code area, my results describe neighbourhoods rather than individual customers. Van der Putten and van Someren (2004) note that customers in the same area can differ, so conclusions about individuals should be drawn with care. As a next step I would rerun the analysis without the duplicates to see how much this choice affects the results.

Reproducibility

The requirements.txt file with the exact package versions was created with the pip freeze command. Danchev (2022, p. 4) explains that listing packages and versions lets others rebuild the same environment and avoids errors from package updates. The README explains how to

create a fresh conda environment with Python 3.12 and how to run the notebook. In Git each step was committed on a dev branch and then pushed to GitHub, so the history shows how the work was built. The notebook runs top to bottom with Run All and the data files are included in the repo.

References

- Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, 5(56), 156. <https://doi.org/10.21105/jose.00156>
- van der Putten, P. (2000). Insurance Company Benchmark (COIL 2000) [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5630S>
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